Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Biology Examination May 2019

Introduction

This document has been prepared for the teachers and candidates of Higher Secondary School Certificate (HSSC-I) Part I Biology. It contains comments on candidates' responses to the 2019 HSSC-I examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the Student Learning Outcomes which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

However, those candidates who did not score well mostly, failed to understand the demand of the question, often missing out important information given in the stimulus or stem. Furthermore, use of scientific terminology and interpretation of unseen diagrams. Nevertheless, candidates outdid in questions without stimulus. However, there is still room for improvement. Mentioned below are few concepts that teachers need to focus so that candidates may perform better.

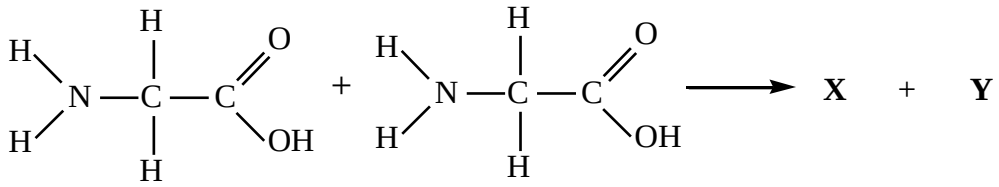
a. Chemiosmosis
b. Illustration of synthesis of linkages
c. Feedback Inhibition
d. Phases and regulation of cardiac cycle
e. Vegetative and floral characteristics of angiospermic families

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1a:

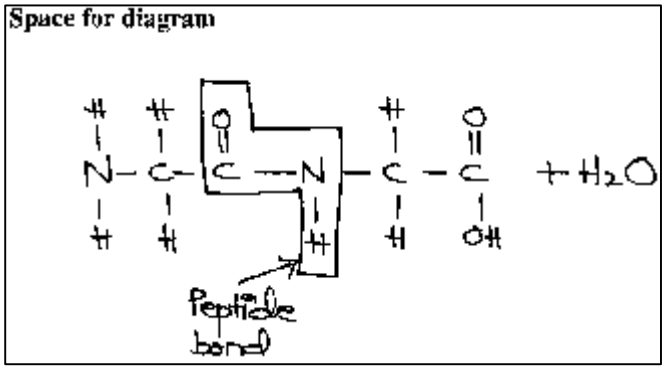
Two amino acid molecules are shown in the given diagram.



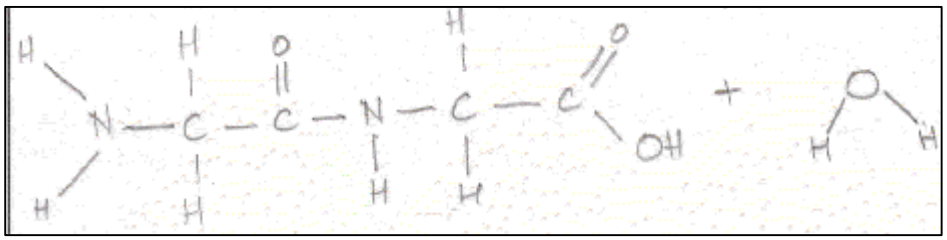
Illustrate the products, X and Y, from the given amino acid molecules.

Better responses reflected wise use of the molecular structures given in the stem, thus, were able to correctly illustrate the products X and Y.

Example 1:



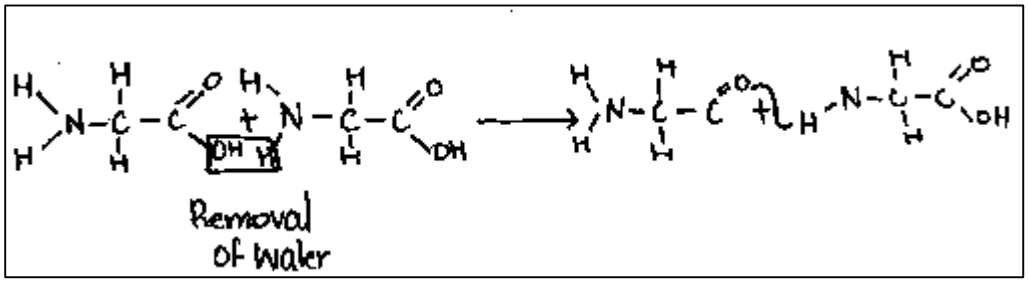
Example 2:



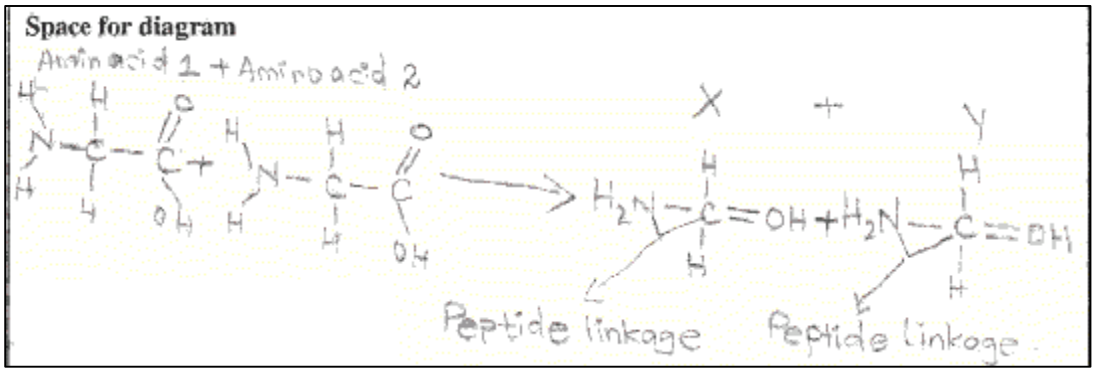
Weaker responses exhibited poor understanding of the illustrations of condensation reaction. Such responses were not able to identify the groups that are involved in the bond formation. In this case, responses showed bond formation between carboxylic group and hydrogen atom. Furthermore, in some of the responses, candidates illustrated the peptide bond without mentioning the removal of water molecule; therefore such responses were not credited marks.

Candidates are advised to focus on the information provided in the stimulus before drafting their responses.

Example 1:

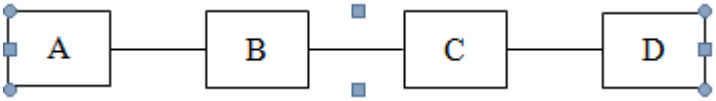


Example 2:



Question 1b:

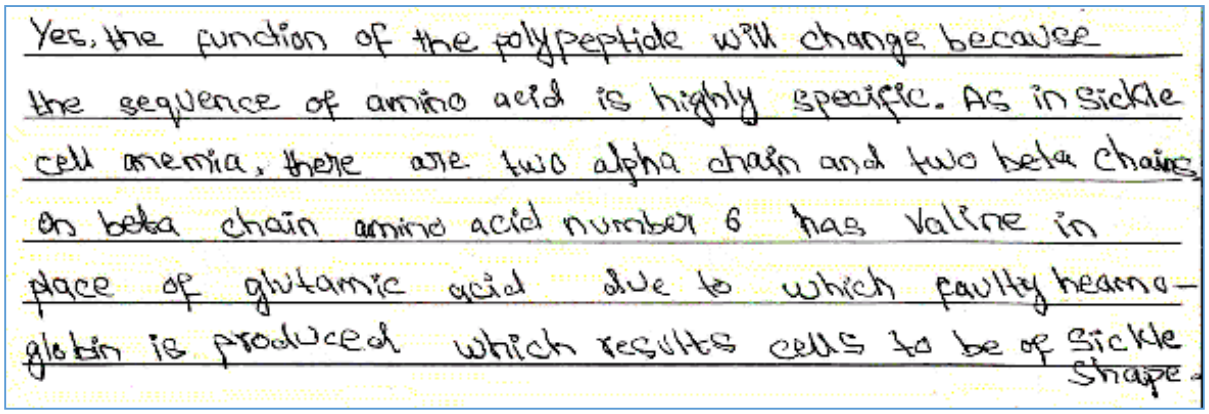
The given diagram represents a sequence of amino acids present in a polypeptide chain.



Will the function of the given polypeptide chain alter, if the amino acid B is replaced with an amino acid H? Elaborate your answer by using the example of sickle cell anaemia.

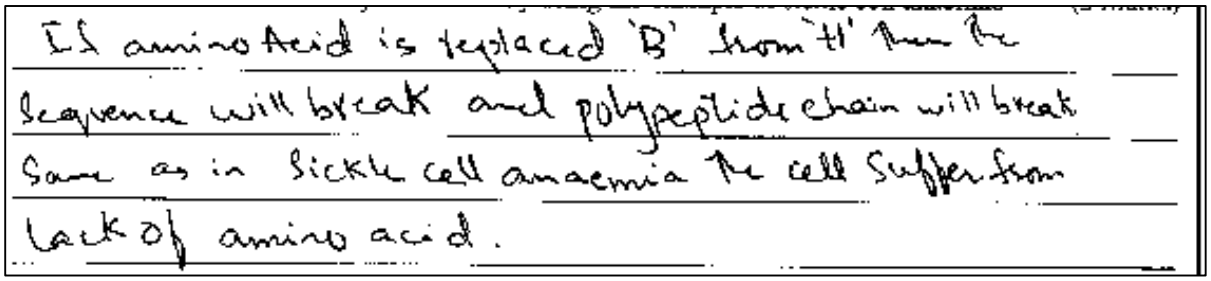
Better responses reflected sound understanding of the concept of significance of sequence of amino acids in a polypeptide chain through sickle cell anaemia. Such responses validated the alteration of amino acid impede function of molecule. This validation is further supported by correct elaboration of sickle cell anaemia; which included the replacement of glutamic acid by valine resulting in alteration in shape and function of red blood cells and function.

Example:

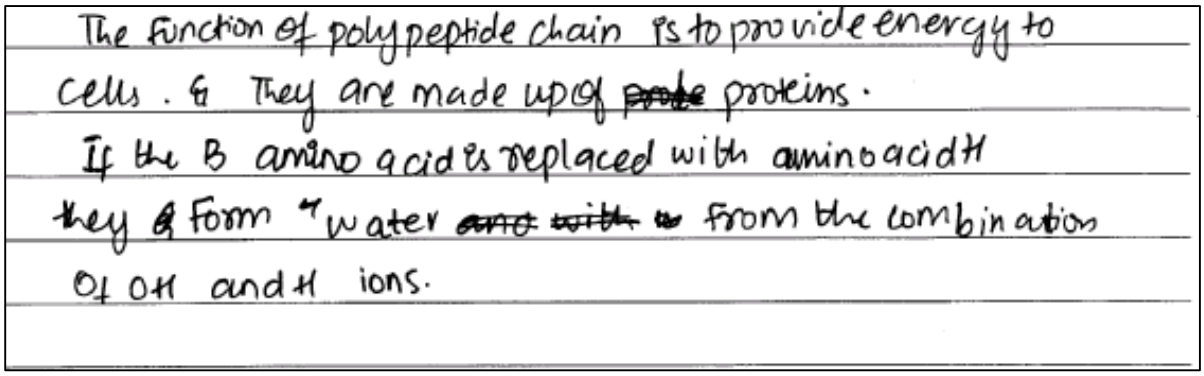

Yes, the function of the polypeptide will change because the sequence of amino acid is highly specific. As in sickle cell anemia, there are two alpha chain and two beta chains on beta chain amino acid number 6 has valine in place of glutamic acid due to which faulty heamo-globin is produced which results cells to be of sickle shape.

Weaker responses displayed candidates' poor understanding of the concept and inability to relate the stimulus with question. In this case, candidates drafted the answers in two ways. Firstly, such responses mentioned that replacement of amino acid B with H will not alter the function of protein and drafted the memorised description of sickle cell anaemia. Secondly, they validated the change in function of protein but could not elaborate it with the example of sickle cell anaemia.

Example 1:

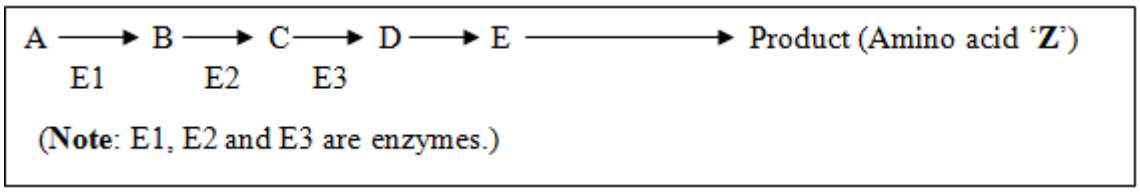

If amino Acid is replaced 'B' from 'H' then the sequence will break and polypeptide chain will break same as in sickle cell anaemia the cell suffer from lack of amino acid.

Example 2:


The function of polypeptide chain is to provide energy to cells. & They are made up of ~~protein~~ proteins.
If the B amino acid is replaced with amino acid H they form "water ~~and with~~ from the combination of OH and H ions.

Question 2:

Given is a biochemical pathway involved in the synthesis of an amino acid '**Z**' in a cell.



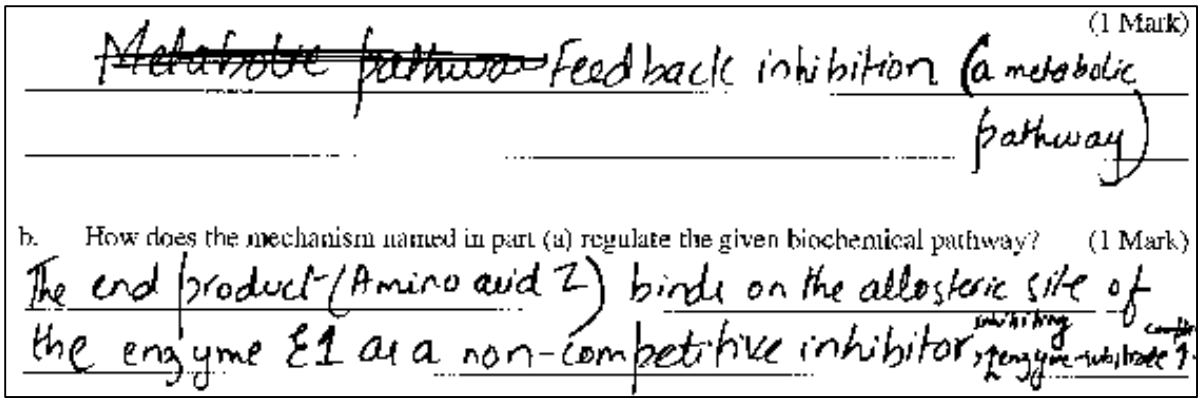
a. Name the mechanism which regulates the amount of amino acid '**Z**' synthesised in the cell.

b. How does the mechanism named in part (a) regulate the given biochemical pathway?

Better responses correctly named the mechanism i.e., negative feedback mechanism/ feedback inhibition in the first part of the question.

In the second part of the question, better responses exhibited candidates' thorough understanding of the concept of negative feedback mechanism. Such responses described the action of product Z given in the stimulus i.e., binding of excess amount of product (Z) at the allosteric site of enzyme 1which results in inhibition of biochemical pathway.

Example 1:

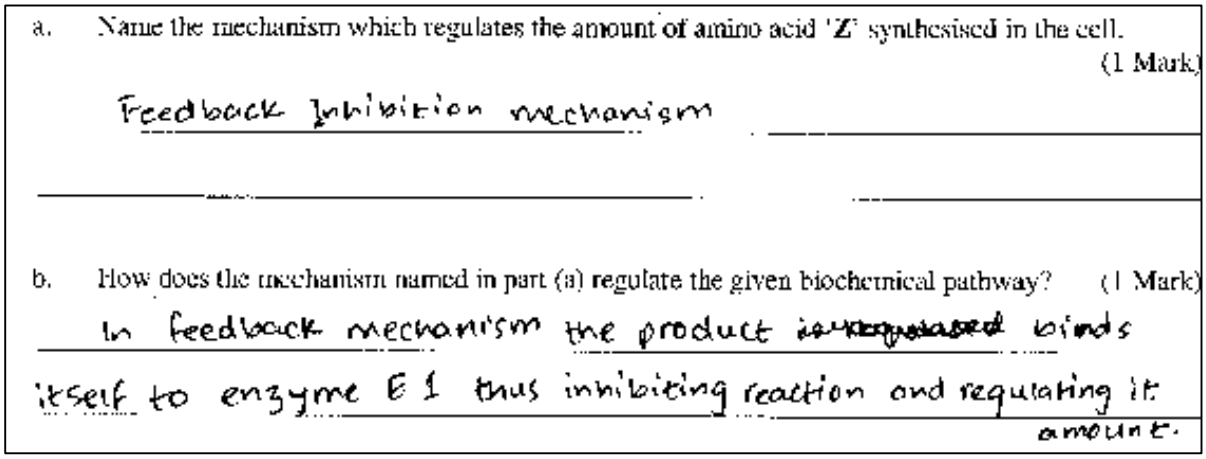

(1 Mark)

~~Metabolic pathway~~ Feed back inhibition (a metabolic pathway)

b. How does the mechanism named in part (a) regulate the given biochemical pathway? (1 Mark)

The end product (Amino acid Z) binds on the allosteric site of the enzyme E1 as a non-competitive inhibitor

Example 2:


a. Name the mechanism which regulates the amount of amino acid 'Z' synthesised in the cell. (1 Mark)

Feedback Inhibition mechanism

b. How does the mechanism named in part (a) regulate the given biochemical pathway? (1 Mark)

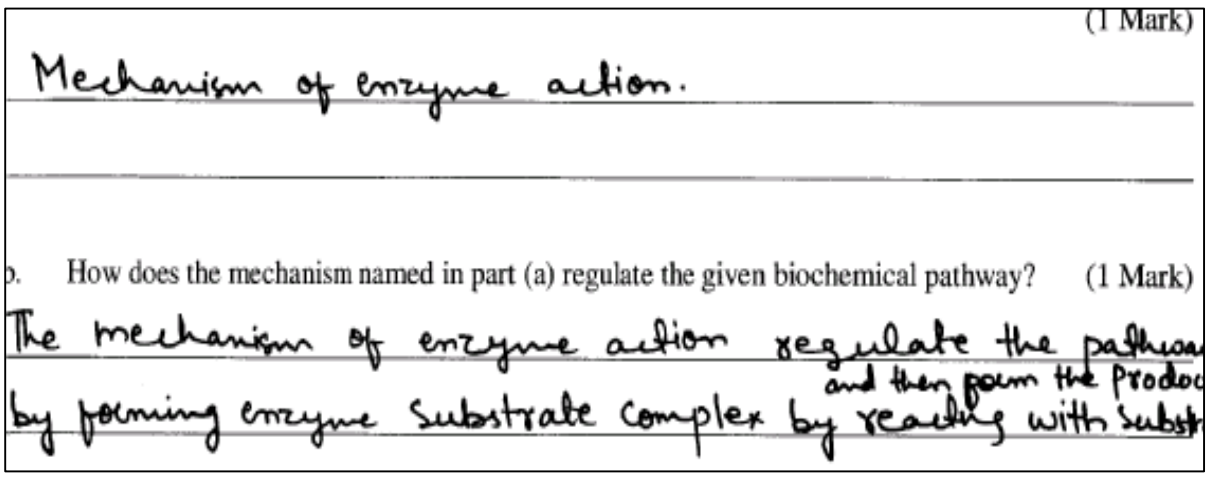
In feedback mechanism the product ~~is~~ binds itself to enzyme E 1 thus inhibiting reaction and regulating it amount.

Weaker responses displayed poor comprehension of the stimulus. In the first part of the question, such responses randomly named the mechanism such as formation of enzyme-substrate complex, mechanism of enzyme action, reversible inhibition and competitive inhibition. However, in some other responses, candidates mentioned the positive feedback mechanism. This shows candidates' misunderstanding of negative and positive feedback mechanism.

In second part of the question, weaker responses mostly described the enzyme-substrate complex formation. Furthermore, some other responses explained the mechanism of enzyme action, i.e., lowering of activation energy. Some other responses provided generalised answers such as binding of inhibitor with enzymes.

For better understanding it is advised to relate this concept with the biochemical pathways given in chapter bioenergetics.

Example:


(1 Mark)

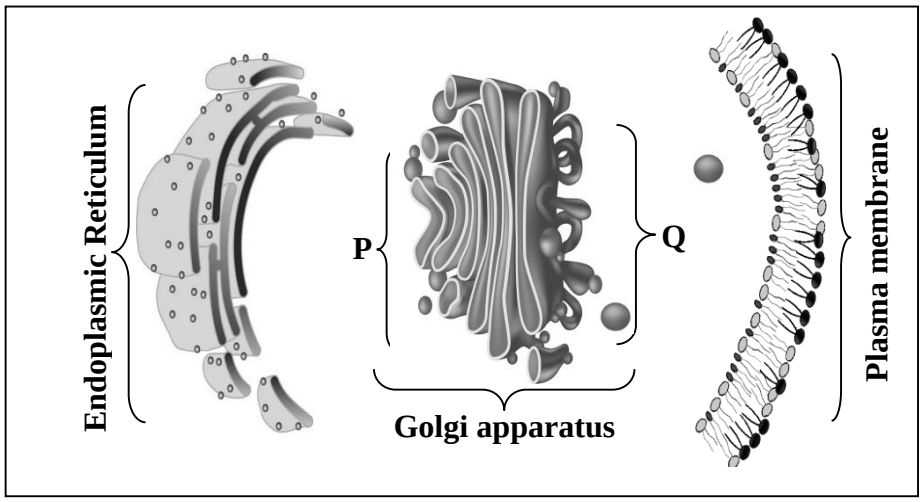
Mechanism of enzyme action.

b. How does the mechanism named in part (a) regulate the given biochemical pathway? (1 Mark)

The mechanism of enzyme action regulate the pathway by forming enzyme substrate complex by reacting with Subst and then form the produc

Question 3:

The given diagram shows three parts of an animal cell.



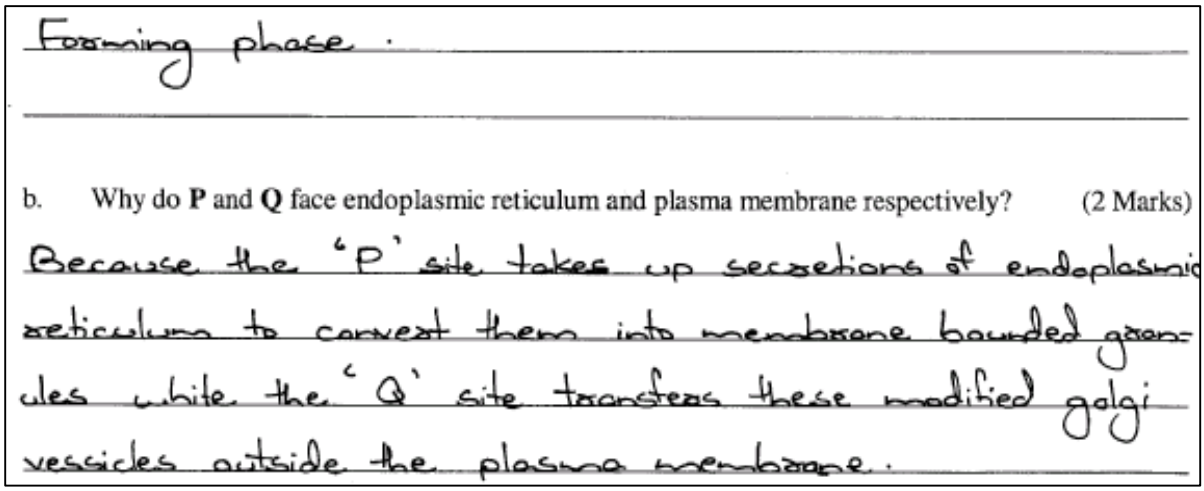
a. Name the labelled side **P** of the Golgi apparatus.

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively?

Better responses displayed clear understanding of the concept of structural orientation of the organelles present in the animal cell. In part 'a' of the question, such responses named the P side as forming site/ synthesis side and cis-face of Golgi apparatus.

In part 'b' of the question, these responses provided the reason of directionality of P and Q. Such responses, drafted as, **P** is directed towards endoplasmic reticulum to receive transport vesicles coming from endoplasmic reticulum. Similarly, **Q** is directed towards plasma membrane to transport material outside of the cell.

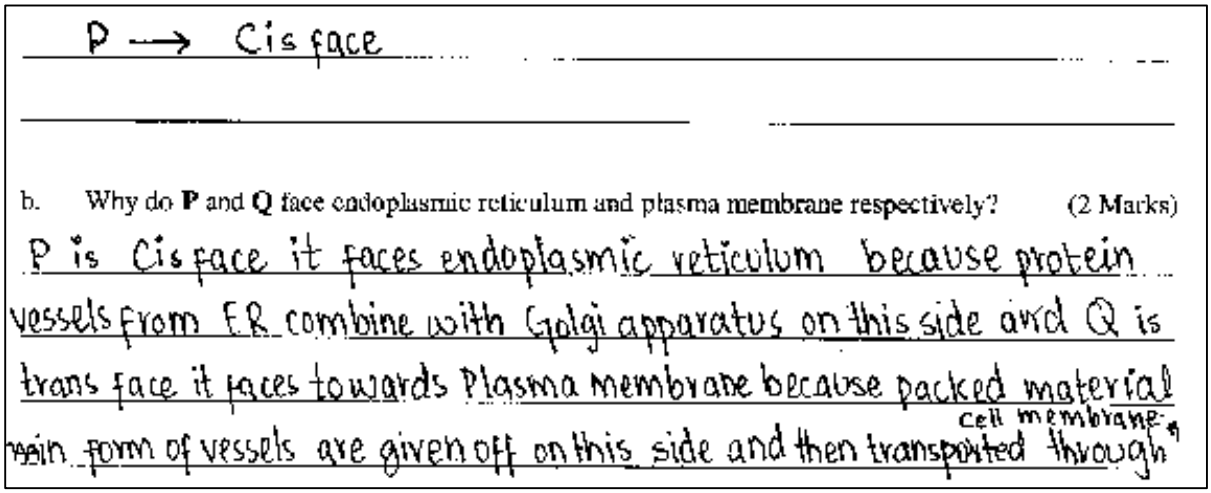
Example 1:


Forming phase.

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively? (2 Marks)

Because the 'P' site takes up secretions of endoplasmic reticulum to convert them into membrane bounded granules while the 'Q' site transfers these modified golgi vessicles outside the plasma membrane.

Example 2:


P → Cis face

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively? (2 Marks)

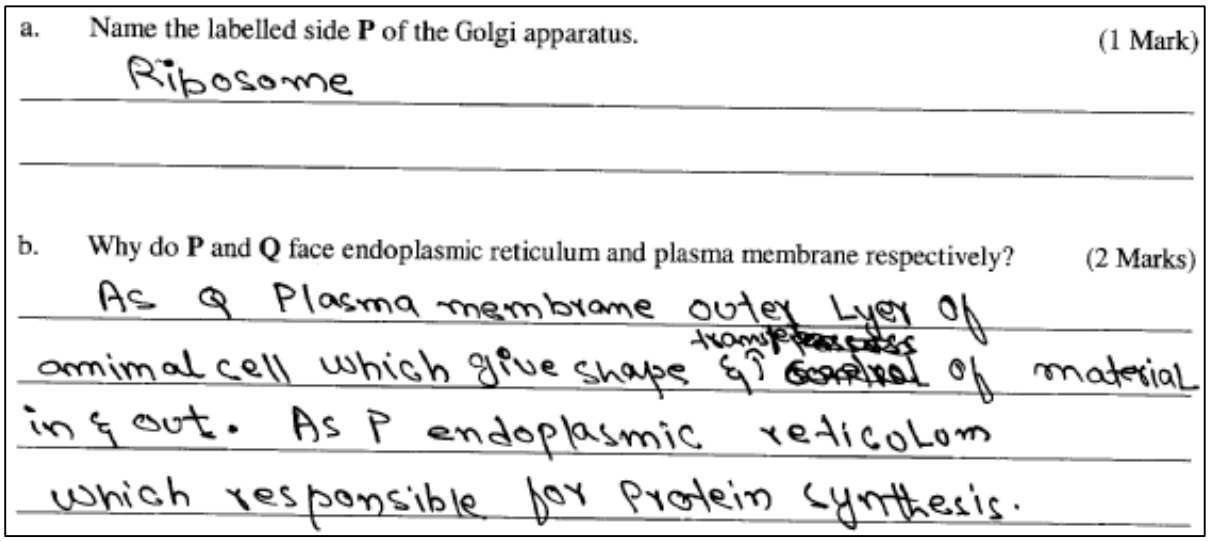
P is Cis face it faces endoplasmic reticulum because protein vessels from E.R combine with Golgi apparatus on this side and Q is trans face it faces towards Plasma membrane because packed material in form of vessels are given off on this side and then transported through cell membrane.

Weaker responses incorrectly named the labelled side in first part of the question. Such response identified as cell wall/ lysosomes/ plasma membrane and microtubules.

In second part of the question, these responses provided vague answers, such as, p is directed to endoplasmic reticulum to function properly and Q is directed towards plasma membrane to provide support and protection.

Teachers are advised to unpack the meaning of 'cis' side of Golgi apparatus which means same side of endoplasmic reticulum. Understanding the meaning of scientific terminologies helps in better understanding of structure and function. Furthermore, it is highly recommended to relate the position, structure and functions of different oragnelles.

Example 1:


a. Name the labelled side **P** of the Golgi apparatus. (1 Mark)

Ribosome

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively? (2 Marks)

As Q Plasma membrane outer Lyer of animal cell which give shape & transport of material in & out. As P endoplasmic reticulum which responsible for Protein synthesis.

Example 2:

a. Name the labelled side **P** of the Golgi apparatus. (1 Mark)

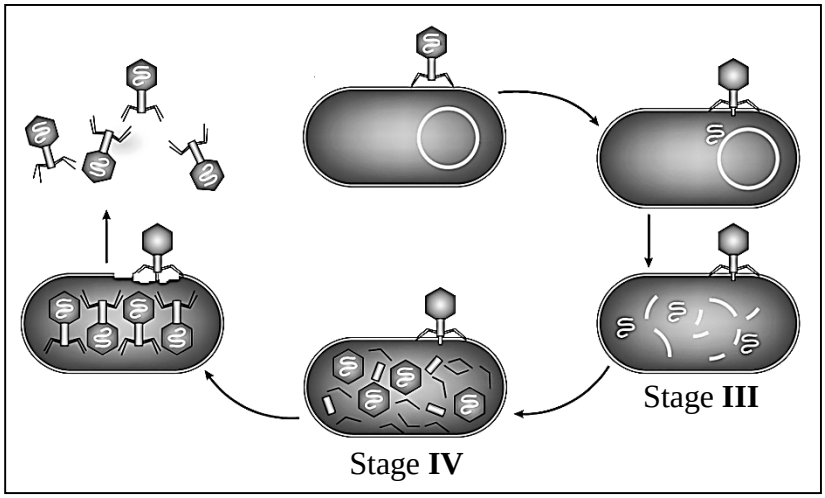
& Cell wall

b. Why do **P** and **Q** face endoplasmic reticulum and plasma membrane respectively? (2 Marks)

Because Golgi apparatus has golgi bodies & their is cell membrane or plasma membrane on one side & rough or smooth endoplasmic reticulum on the another side.

Question 4:

The given diagram shows the life cycle of a bacteriophage.



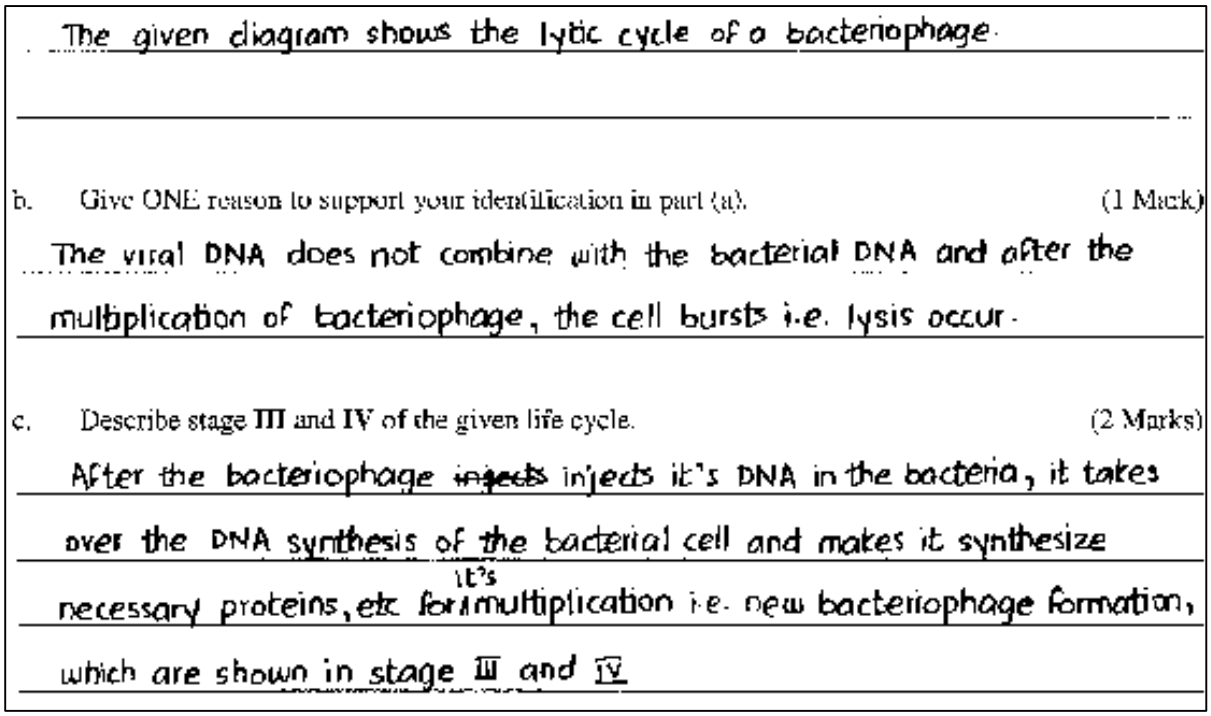
a. Identify the type of life cycle shown in the given diagram.

b. Give ONE reason to support your identification in part (a).

c. Describe stage **III** and **IV** of the given life cycle.

Better responses wisely inferred the diagram to identify the life cycle of bacteriophage, i.e. lytic cycle in the part 'a' of the question. Similarly, in part 'b' of the question, these responses provided correct reason of their identification by mentioning lysis of the bacterial cell at the end of cycle. In part 'c' of the question, these responses described the stage III by highlighting the control of host's bio-synthetic machinery by bacteriophage, likewise, in stage IV, synthesis of viral components such as DNA, proteins, head and capsid is included in their description.

Example:



The given diagram shows the lytic cycle of a bacteriophage.

b. Give ONE reason to support your identification in part (a). (1 Mark)

The viral DNA does not combine with the bacterial DNA and after the multiplication of bacteriophage, the cell bursts i.e. lysis occur.

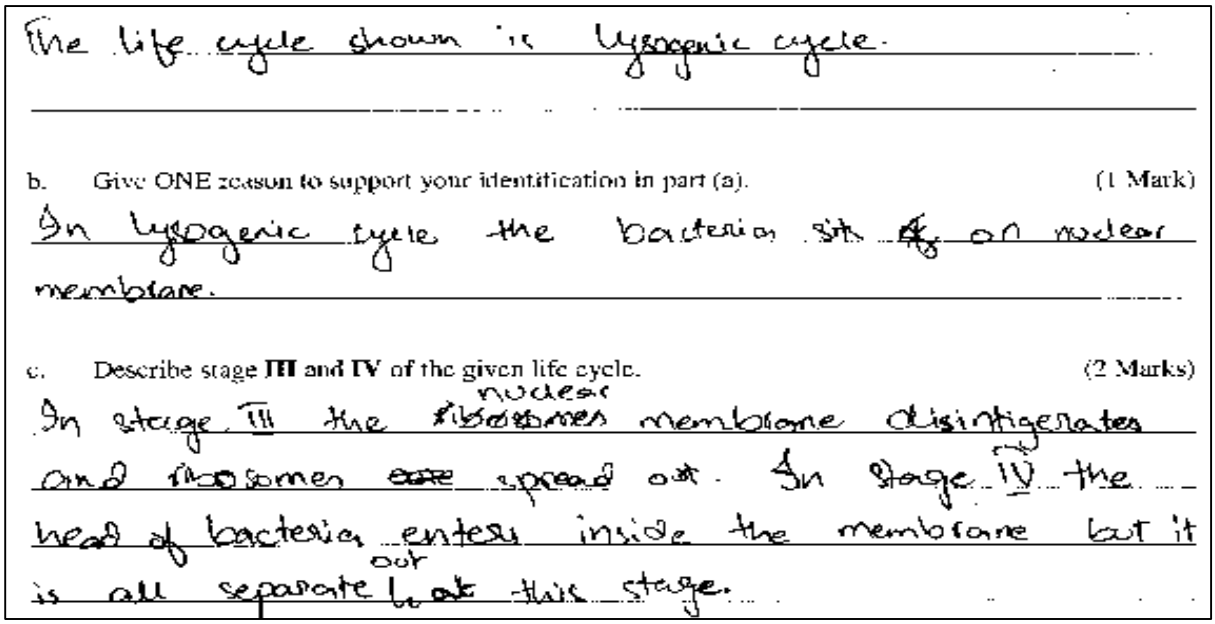
c. Describe stage III and IV of the given life cycle. (2 Marks)

After the bacteriophage ~~injects~~ injects it's DNA in the bacteria, it takes over the DNA synthesis of the bacterial cell and makes it synthesize necessary proteins, etc for (it's) multiplication i.e. new bacteriophage formation, which are shown in stage III and IV

Weaker responses incorrectly identified the life cycle of bacteriophage, therefore, provided irrelevant answers to the rest of the questions. Such responses identified the given life cycle as lysogenic in part 'a' of the question, therefore, failed to provide correct reason in part 'b' of the question. These responses mentioned ambiguous answers like attachment of bacteriophage with the host cell wall/ injection of viral DNA into host cell. In part 'c' of the question, such responses provided vague description of each stage. Initiation of replication viral DNA and insertion of viral DNA into the host cell DNA in stage III and IV respectively were frequently mentioned. Hence, these responses reveal a significant confusion between lysogenic and lytic cycle of bacteriophage.

Teachers are advised to provide worksheet having missing steps of each of lysogenic and lytic cycle for more practice.

Example:


The life cycle shown is Lysogenic cycle.

b. Give ONE reason to support your identification in part (a). (1 Mark)

In lysogenic cycle the bacteria sits on nuclear membrane.

c. Describe stage **III** and **IV** of the given life cycle. (2 Marks)

In stage III the nuclear membrane disintegrates and ribosomes spread out. In stage IV the head of bacteria enters inside the membrane but it is all separate out at this stage.

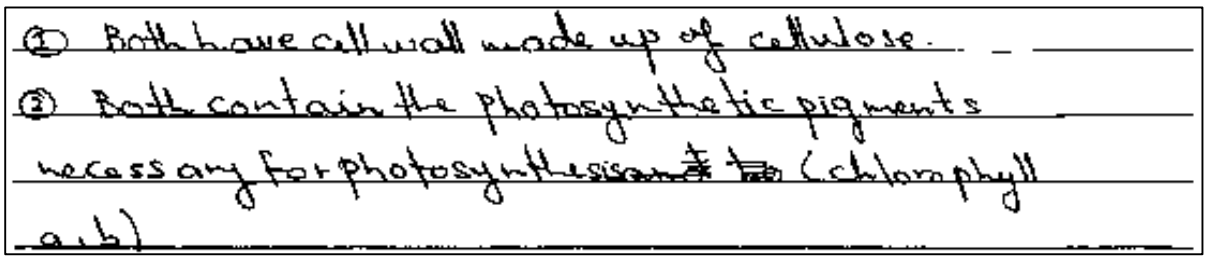
Question 5a:

Plants have evolved from green algae.

Mention any TWO evidences of this evolution.

Better responses correctly mentioned evidences of evolution such as presence of photosynthetic pigments chlorophyll a and b and carotenoids in plants and green algae, /presence of similar energy reserve in the form of starch/ presence of similar cell wall/ presence of photosystem I and II/and showing similar RNA sequencing.

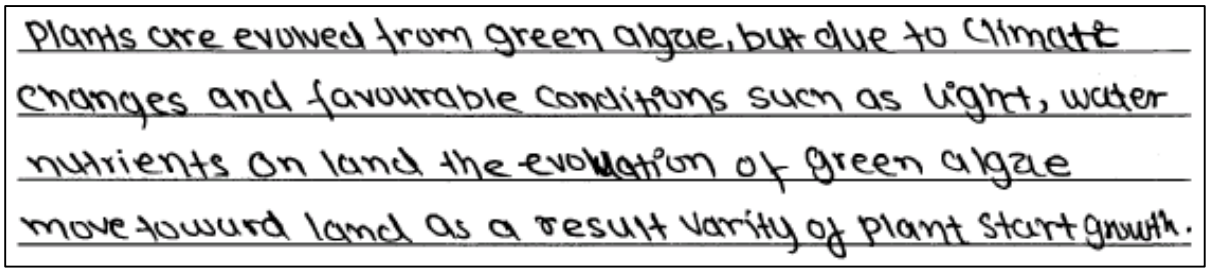
Example:


① Both have cell wall made up of cellulose.

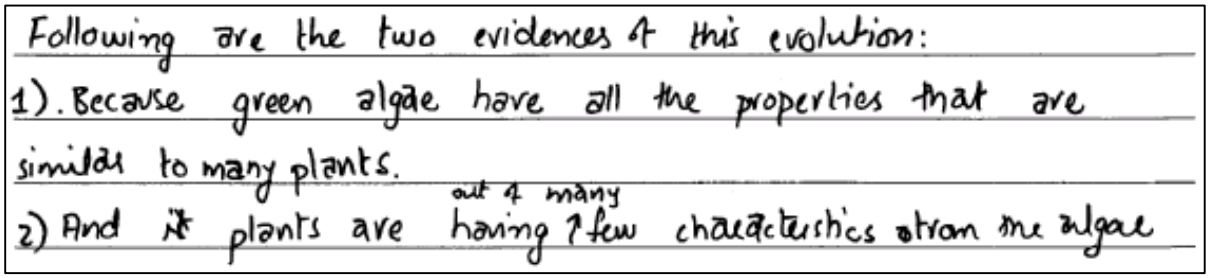
② Both contain the photosynthetic pigments necessary for photosynthesis (chlorophyll a,b)

Weaker responses exhibited ambiguity in the concept of evolution of plants from green algae. These responses mentioned general features of plants and green algae. Moreover, few of the responses mentioned distinguishing characteristics of plants and green algae. Such as plant body is differentiated into root, stem and leaves, whereas, green algae has thallus body/ plants have vascular bundles, whereas, green algae do not possess. However, some candidates mentioned the placement of green algae in the kingdom Protista.

Example 1:


Plants are evolved from green algae, but due to climate changes and favourable conditions such as light, water nutrients on land the evolution of green algae move toward land as a result varity of plant start growth.

Example 2:

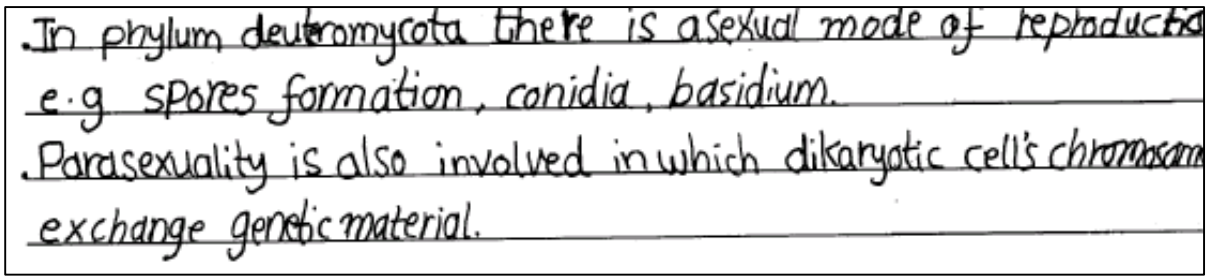

Following are the two evidences of this evolution:
1). Because green algae have all the properties that are similar to many plants.
2) And if plants are out of many having few characteristics from the algae

Question 5b:

Describe the mode of reproduction of phylum deuteromycota of fungi.

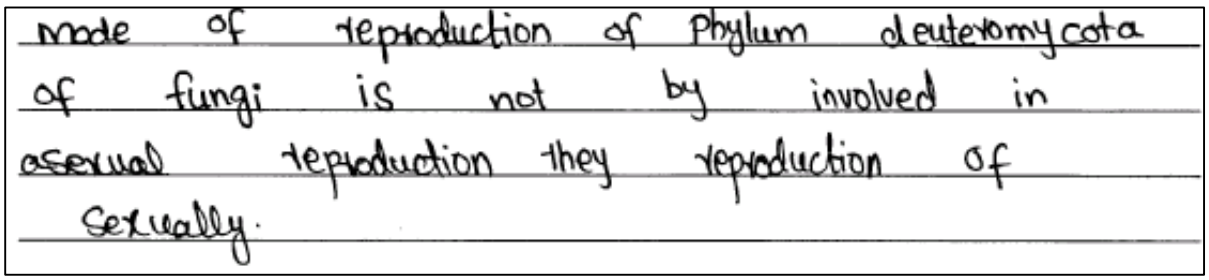
Better responses correctly described the parasexual mode of reproduction of phylum deuteromycota. Such responses highlighted exchange of portions of chromosomes of two nuclei lying in the same hypha. Moreover, other credit worthy responses described asexual mode of reproduction through conidia and spore formation.

Example:


.In phylum deuteromycota there is asexual mode of reproductio
e.g spores formation, conidia, basidium.
.Parasexuality is also involved in which dikaryotic cell's chromosom
exchange genetic material.

Weaker responses revealed candidates' poor concept of imperfect fungi i.e. deutromycota. These responses mentioned mode of reproduction as sexual and asexual. However, some responses mentioned the asexual reproduction with incorrect description i.e. budding and binary fission. Therefore these responses were unable to obtain full marks.

Example:

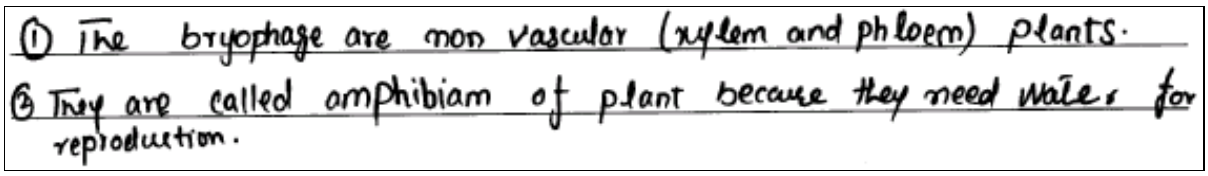

mode of reproduction of Phylum deuteromycota of fungi is not by involved in asexual reproduction they reproduction of Sexually.

Question 6a:

State any TWO characteristic features of bryophytes.

Better responses exhibited thorough understanding of the key terms, i.e. 'characteristic features', therefore, were able to state features such as absence of vascular bundles/ dominant gametophyte/ attachment of sporophyte to gametophyte/ homospory/, seta and capsule/ confined to damp shady places (dependant on water for reproduction)/ most of the body is parenchymatous/ amphibious in nature/ possesse psuedo stem and leaves/ plant body is gametophtye / presence of rhizoids for water absoprtion/ heteromorphic alternation of generation.

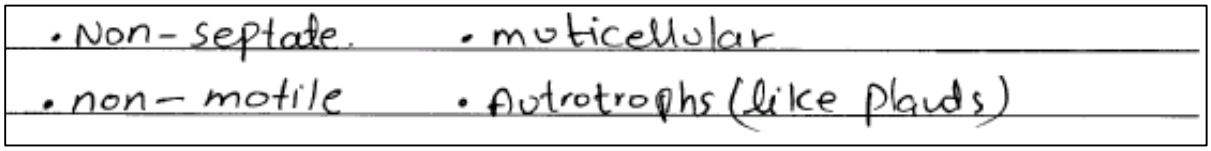
Example:


① The bryophage are non vascular (xylem and phloem) plants.
② They are called amphibiam of plant because they need wate, for reproduction.

Weaker responses showed apparent lack of understanding of the key term given in the question that led to irrelevant answers. Such responses 'stated' the general features rather than the 'characteristics' features. Majority of these responses stated multicellularity and autorophic/ photosynthetic nature as the characteristic feature. Some other responses stated the memorised features of fungi, tracheophtyes and sub-phylum of the Kingdom Plantae.

Candidates are advised to identify and understand the key word(s) given in the question.

Example:


• Non-septate. • muticellular
• non-motile • autrotrophs (like plauds)

Question 6b:

Differentiate between Solanacea (potato family) and Fabaceae (pea family) with reference to their floral characteristics in the given table.

Floral Characteristic	Solanacea	Fabaceae
Inflorescence		
Placentation		
Flower Symmetry		

Better responses manifested comprehensive understanding of the terms, i.e. inflorescene, placentation and flower symmetry. These responses clearly differentiated the characteristics of solanacea and fabaceae by differentiating the inflorescence on the basis of arrangement of flowers on the shoot such as axillary cyme or combination of cymes/ helicoid/ axillary umbellate cyme and racemose/ solitary axillary respectively. Similarly, differentiated placentation on the basis of ovules attached to the wall of ovary such as axile and marginal placentation in solanacea and fabacea respectively. Likewise, flower symmetry is differentiated on the basis of arrangement of floral leaves, i.e. actinomorphic/ regular/ radial symmetry or weakly zygomorphic and zygomorphic/ bilateral symmetry in slonacea and fabacea respectively. Those responses which mentioned the correct floral formula were also credit marks.

Example:

Floral Characteristic	Solanacea	Fabaceae
Inflorescence	cyme.	Recyme
Placentation	Axillary placentation	Marginal placentation.
Flower Symmetry	Actinomorphic/Zygomorphic	Zygomorphic

Weaker responses displayed poor understanding of the terms, therefore failed to differentiate correctly. Majority of these responses differentiated on the basis of vegetative characteristics, such as, habitat, root and stem conditions of solanacea and fabaceea were given. Few responses struggled to provide the floral formula. Moreover, some responses differentiated on the basis of economic importance. However, some of the candidates swapped the differentiating points of solanacea and fabacea.

Teachers are advised to integrate the related topic given in the practical and theory syllabus for better understanding. Moreover, facilitate the candidates with worksheets having explanation of each term with the diagram.

Example 1:

Floral Characteristic	Solanacea	Fabaceae
Inflorescence	⊕ ♀ $C_5 A_5 G_2$	⊕ ♀ $C_{(5)} A_5 G_1$
Placentation	Grown underground with fined roots and stem without seed.	Grows on pea plant with seed.
Flower Symmetry	They do not have seeds and have buds on body.	The have seeds and are enclosed in covering.

Example 2:

Floral Characteristic	Solanacea	Fabaceae
Inflorescence	non-flowering	non-flowering
Placentation	Budding	Seed dispersal
Flower Symmetry	Bi-lateral	Bi-lateral

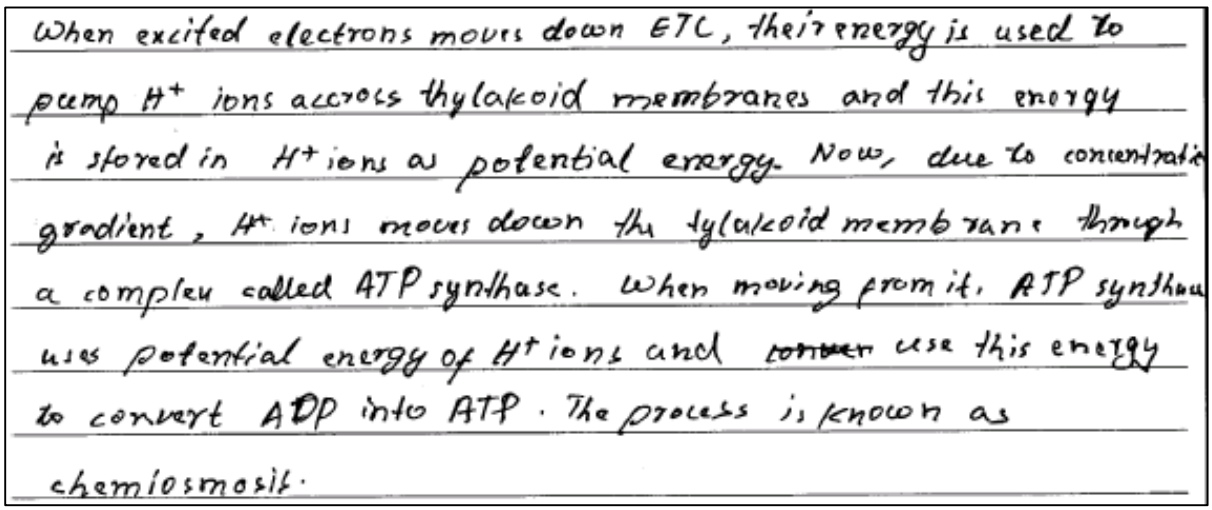
Question 7:

How does electron transport chain facilitate the production of adenosine tri phosphate (ATP) through chemiosmosis?

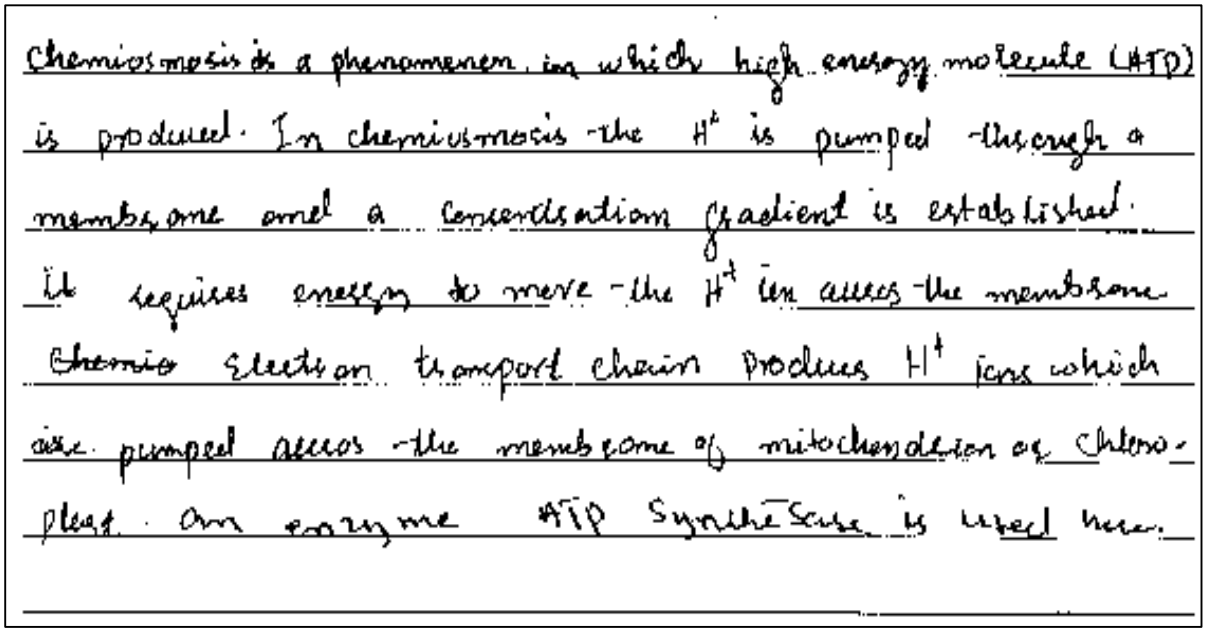
Better responses clearly described the mechanism through which electron transport chain utilises the proton gradient to produce ATP. Such responses described that NADH and $FADH_2$ carry high energy electrons from the glycolysis and Krebs cycle to the electron transport chain. These electrons are pulled along electron transport chain they release energy that is used to pump protons across the cristae membrane, and this creates a proton gradient. This gradient is stored or potential energy. Chemiosmosis uses the potential energy in the form of proton gradient to convert ADP to ATP. For this a special complex is used called ATP synthase. During the diffusion of protons through the ATP synthase electrons are used to make ATP.

Those responses which provided correct description of ATP formation through electron transport chain in photosynthesis and respiration were credited full marks.

Example 1:


When excited electrons moves down ETC, their energy is used to pump H^+ ions accross thylakoid membranes and this energy is stored in H^+ ions as potential energy. Now, due to concentration gradient, H^+ ions moves down the thylakoid membrane through a complex called ATP synthase. When moving from it, ATP synthase uses potential energy of H^+ ions and ~~convert~~ use this energy to convert ADP into ATP. The process is known as chemiosmosis.

Example 2:


Chemiosmosis is a phenomenen in which high energy molecule (ATP) is produced. In chemiosmosis the H^+ is pumped through a membrane and a concentration gradient is established. It requires energy to move the H^+ ion across the membrane. Chemio Electron transport chain produces H^+ ions which are pumped across the membrane of mitochondrion or Chloroplast. An enzyme ATP Synthase is used here.

Weaker responses exhibited confusion in phosphorylation and electron transport chain. Most frequently appearing weak answer was photolysis of water and use of electrons in electron transport chain. Furthermore, few responses described the reactions of Calvin cycle. Other responses provided very brief description of meaning of chemiosmosis.

Example 1:

Electron transport chain which help for the process of ATP it is because the electron transport chain provide the chemical for making their food It is because when the rays of Sunlight are not available for the process of Photosythesis So the chemosythesis help for making their for by the help of chemicals

Example 2:

when $NADPH^+$ are gain hydrogen and split 2 electron than they become NADPH and at the same time adenosine triphosplati is formed by Chemiosmosis.

Question 8:

a.

Name the secretory cells which are common to both, small intestine and large intestine.

b.

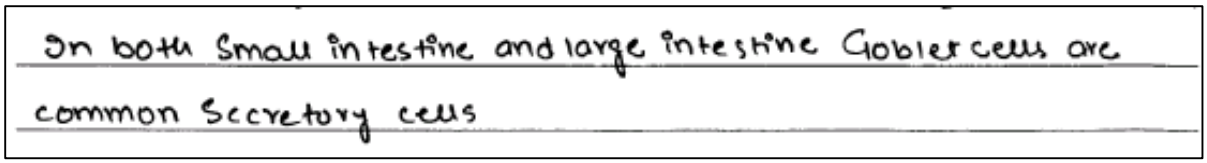
Differentiate between the large intestine and small intestine of human beings on the basis of their structure.

S. No.	Small Intestine	Large Intestine
i.		
ii.		
iii.		
iv.		

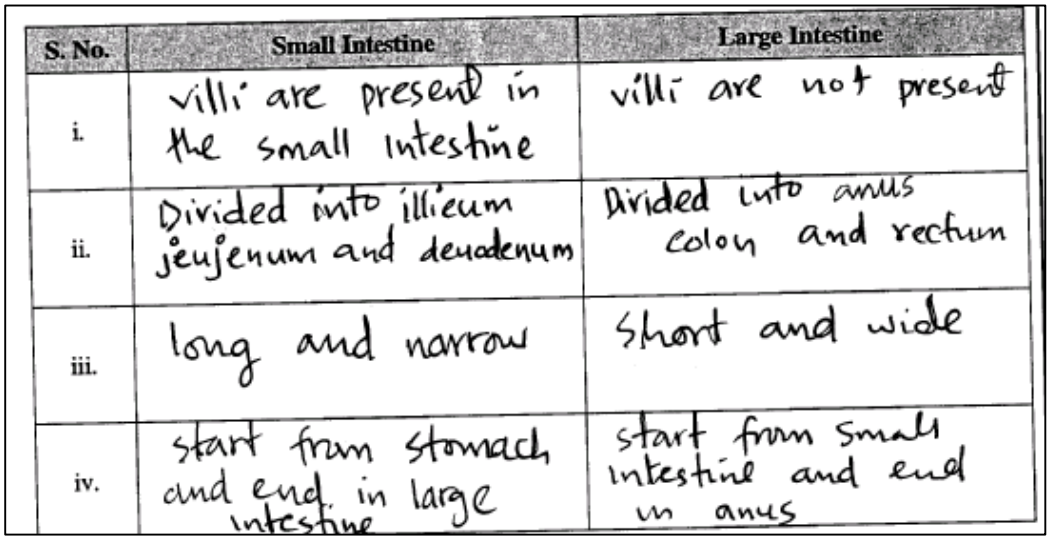
Better responses correctly identified the secretory cells by naming goblet cells/ mucous secreting cells in part 'a' of the question.

In part 'b' of the question, better responses showed thorough understanding of the anatomy of small intestine and large intestine. These responses mentioned difference in number of divisions, length, size of lumen, presence of specialized structures for example villi and internal circular folds and overall structure of small and large intestine.

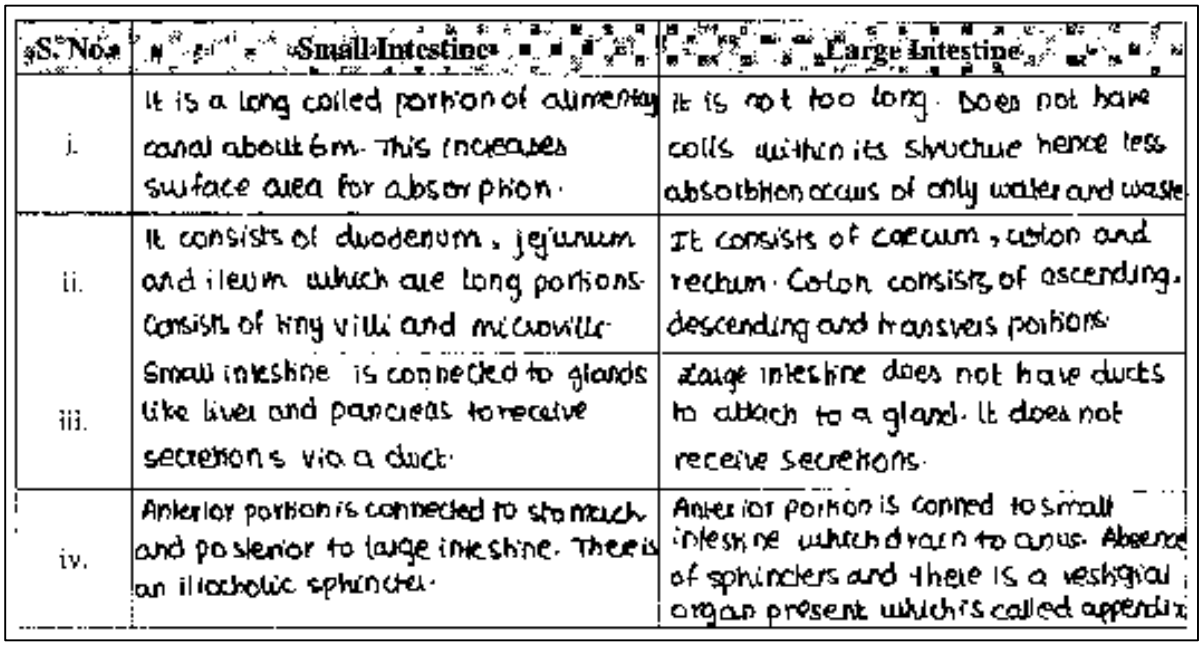
Example 8a:



Example 8b:



S. No.	Small Intestine	Large Intestine
i.	villi are present in the small intestine	villi are not present
ii.	Divided into illieum jeujenum and deuadenum	Divided into anus colon and rectum
iii.	long and narrow	short and wide
iv.	start from stomach and end in large intestine	start from small intestine and end in anus



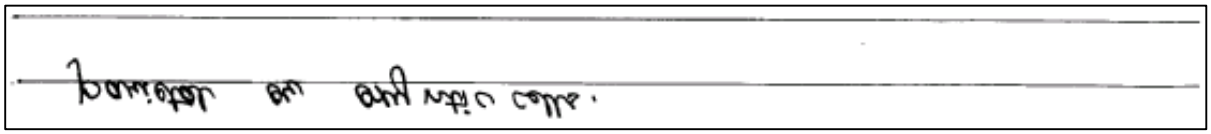
S. No.	Small Intestine	Large Intestine
i.	It is a long coiled portion of alimentary canal about 6m. This increases surface area for absorption.	It is not too long. Does not have coils within its structure hence less absorbtion occurs of only water and waste.
ii.	It consists of duodenum, jejunum and ileum which are long portions. Consists of tiny villi and microvilli.	It consists of caecum, colon and rectum. Colon consists of ascending, descending and transverse portions.
iii.	Small intestine is connected to glands like liver and pancreas to receive secretions via a duct.	Large intestine does not have ducts to attach to a gland. It does not receive secretions.
iv.	Anterior portion is connected to stomach and posterior to large intestine. There is an ileocholic sphincter.	Anterior portion is conned to small intestine which drain to anus. Absence of sphincters and there is a veshigial organ present which is called appendix

Weaker responses exhibited candidates' poor understanding of nuts and bolts of comparative anatomy and histology of small and large intestine.

In part 'a' of the question, these responses mentioned all the memorised names of secretory cells related to digestive system. For example, the frequently written names were zymogen cells/ oxyntic cells and parietal cells.

In part 'b' of the question, majority of these responses differentiated small and large intestine on the basis their function. Such as dipeptide, maltose and fatty acids are converted into smaller subunits in small intestine whereas large intestine is not involve in digestion/ absorption of materials takes place in small intestine whereas, in large intestine reabsorption of water takes place.

Example 8a:



Example 8b:

S. No.	Small Intestine	Large Intestine
i.	It consist of duodenum, jejunum and ileum.	It consist of pancreas and bile duct.
ii.	It is connected to stomach.	whereas it is connected to excretory organ.
iii.	It is ~~folding~~ folded tube.	It is not folded tube.
iv.	It helps in the secretion of food.	It helps in secretion as well in excretory products to go in excretory organ.

Question 9:

Complete the given table for tuberculosis.

Cause (Name of Bacteria)	Symptom
	i.
	ii.

Better responses displayed complete knowledge of causative agent and symptoms of tuberculosis by stating the correct name of causative agent i.e., *Mycobacterium tuberculosis* and symptoms such as cough/ shortness of breath/ weight loss and septum with bleeding.

Example:

Cause (Name of Bacteria)	Symptom
mycobacterium tuberculosis.	i. weight loss
	ii. chills, night sweat.

Weaker responses provided either incorrect or incomplete name of causative agent of tuberculosis. Furthermore, these responses stated irrelevant symptoms such as diarrhoea/ abdominal pain/ dry mouth/ patches on body and mood swings. However, some responses stated generalised symptoms like fever and weakness were awarded marks.

Example:

Cause (Name of Bacteria)	Symptom
Pathogens	i. Clots in the passage of blood flow.
	ii. Damage blood vesels. Pain in vessels.

Extended Response Questions (ERQs)

The following questions (10 and 11) offered a choice between part **a** and **b**.

Approximately, larger number of candidates attempted part 'a' of question 10. This shows their interest in the concept of 'plasma membrane' than part 'b', i.e. 'evolution of vertebrates and their adaptive features'. Similarly, in question 11, more preference of candidates was observed in part 'b' than part 'a'. This indicates that candidates are more confident in the concept of cardiac cycle than respiratory system.

Question 10a:

i. Describe the structure attained by phospholipid molecules when they are placed in water.

ii. Describe any FIVE functions of the phospholipid membrane (plasma membrane).

Better responses in first part of the question, revealed thorough understanding of chemistry of phospholipid molecule and its structural adaptability in different conditions. These responses described the formation of micelles such that when phospholipid molecules are placed in water, the lipid ends (hydrophobic) of molecules tend to aggregate forming micelles with a phosphate group projecting out into the water. Further, these molecules form a bilayer with the polar ends facing water and non-polar ends facing each other.

In second part of question, these responses obtained full marks by providing complete description of role of phospholipid in terms of protection of cell against pathogens/ transport of materials in and out of cell through different means like phagocytosis, pinocytosis, transport protein, gated channels and exocytosis/ conduction of nerve impulse in nerve cells/ regulation of cellular composition/ maintenance of cell shape and determination cell identity.

Example:

(i) When phospholipid molecule is placed in H_2O

As phospholipid molecule is non-polar, when placed in water, it is become ring shaped having a long hydrophobic tail and circular shaped hydrophilic Phosphate containing head. It remains insoluble.

(ii) Functions of plasma membrane

As the following functions are carried by plasma membrane due to its complex structure.

(1) The main function of plasma membrane is to regulate the entry and exit of substance between cell and surrounding.

(2) Secondly, the plasma membrane also pinches inward forming food vacuole called endocytosis, endocytosis may be taking in solid material (Phagocytosis) and pinocytosis of intaking food in form of Liquid.

(3) The most important function of plasma membrane is that, it maintains a steep concentration gradient which helps in exchange of cellular materials in and out of cells.

(4) It is selectively or differentially permiable membranes it means that it functions to prevent the entry of wasteful and harmful substances that affect cell and damage them.

(5) In some bacteria, cell membrane also contain respiratory enzymes that helps the to carry out cellular respiration and synthesize ATP which is cell's energy currency.

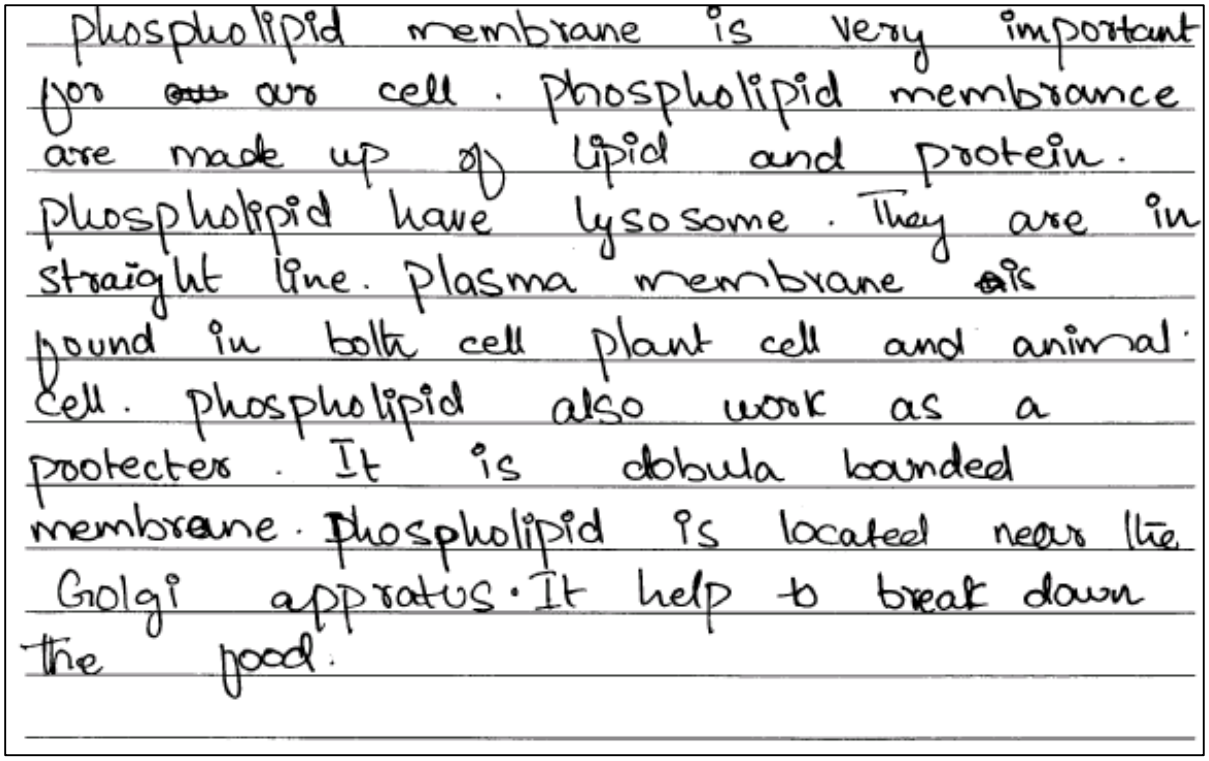
(6) Phospholipid membrane also encloses organelles like mitochondria, golgi bodies, ribosomes e.t.c In other words it encloses cytosol which contains materials of life.

Thus without the phospholipid membrane, it is suggested that cell never stay alive at it has buches of functions it serves for cell.

Weaker responses were unable to describe the condition of phospholipid molecule when placed in water in first part of question. However, some responses described the general structure of phospholipid molecules but failed to articulate in the condition asked. Some responses drafted the chemical composition of different classes of lipid such as composition of triglycerides/ waxes/ saturated and unsaturated fatty acids.

In second part of the question, revealed candidates' limited knowledge and understanding about the functions of phospholipid membrane. Majority of these responses described semi-permeability of membrane and protection of cell only. Moreover, few responses elaborated the composition of the phospholipids. Additionally, these responses mentioned the functions of other organelles of cell such as functions of Golgi apparatus and mitochondria were frequently given.
Teachers are advised to categorise functions of phospholipid membrane while teaching the topic.

Example:


phospholipid membrane is very important for our cell. phospholipid membrance are made up of lipid and protein. phospholipid have lysosome. They are in straight line. plasma membrane is found in bolh cell plant cell and animal cell. phospholipid also work as a protecter. It is dobula bounded membrane. phospholipid is located near the Golgi apparatus. It help to breat down the food.

Question 10b:

Reptiles have evolved to replace amphibians as dominant terrestrial animals.

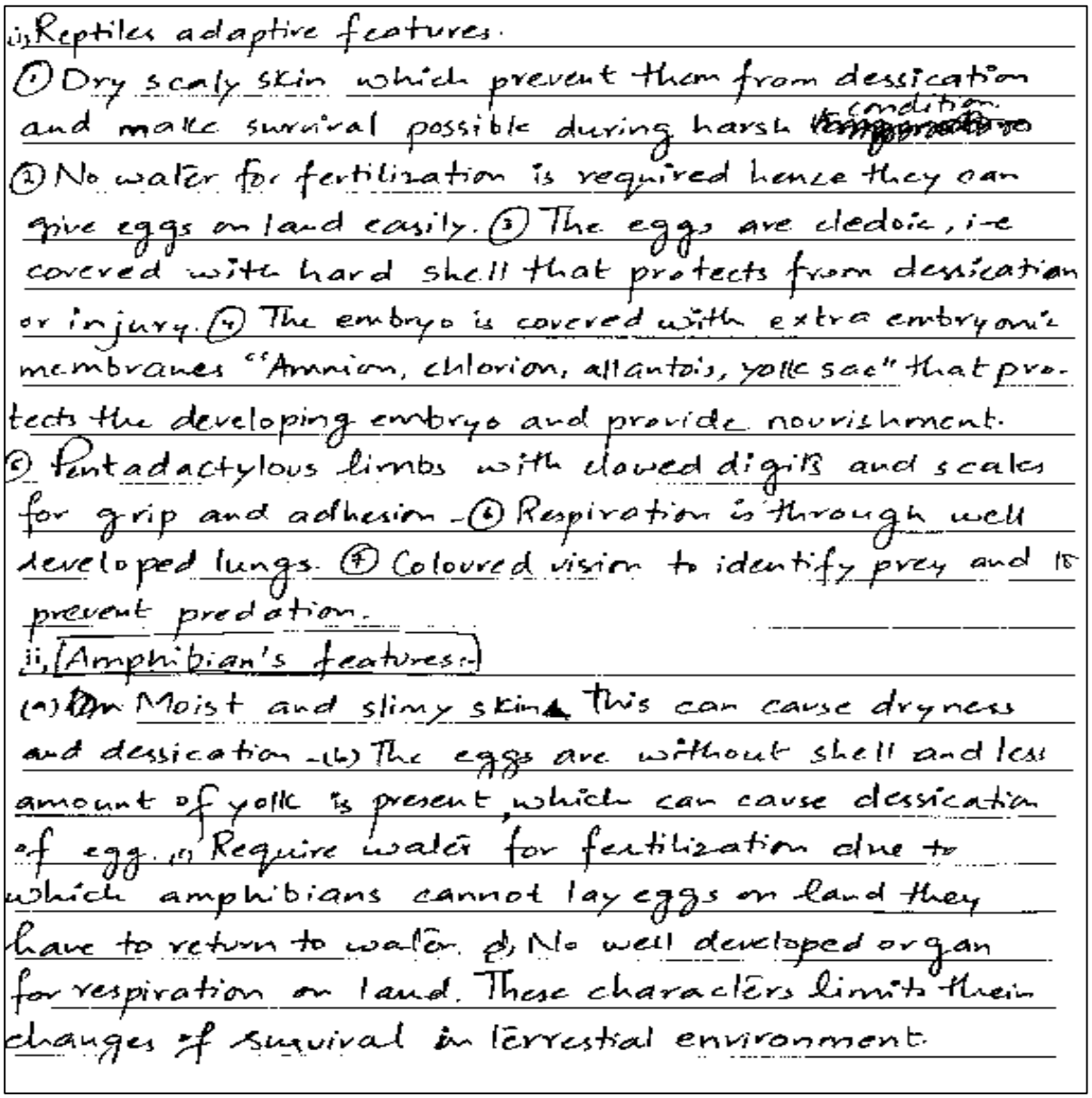
i. Describe any FIVE adaptive features of reptile to the terrestrial environment.

ii. Describe any TWO features of amphibians which limit their chances of survival in the terrestrial environment.

Better responses revealed sound understanding of evolution of reptiles by describing the adaptive features of reptiles such as development of copulatory organ for internal mode of fertilisation/ development of amniotic egg/ shelled egged/ presence of dry scaly skin to prevent them from dissection and limbs for efficient locomotion.

Similarly, in second part of the question, these responses mentioned the features of amphibian which limit their chances of survival on land are permeable and moist skin/ external mode of fertilisation/ unprotected eggs/ eggs having insufficient amount of yolk/ lack of amnion and incomplete metamorphosis.

Example:



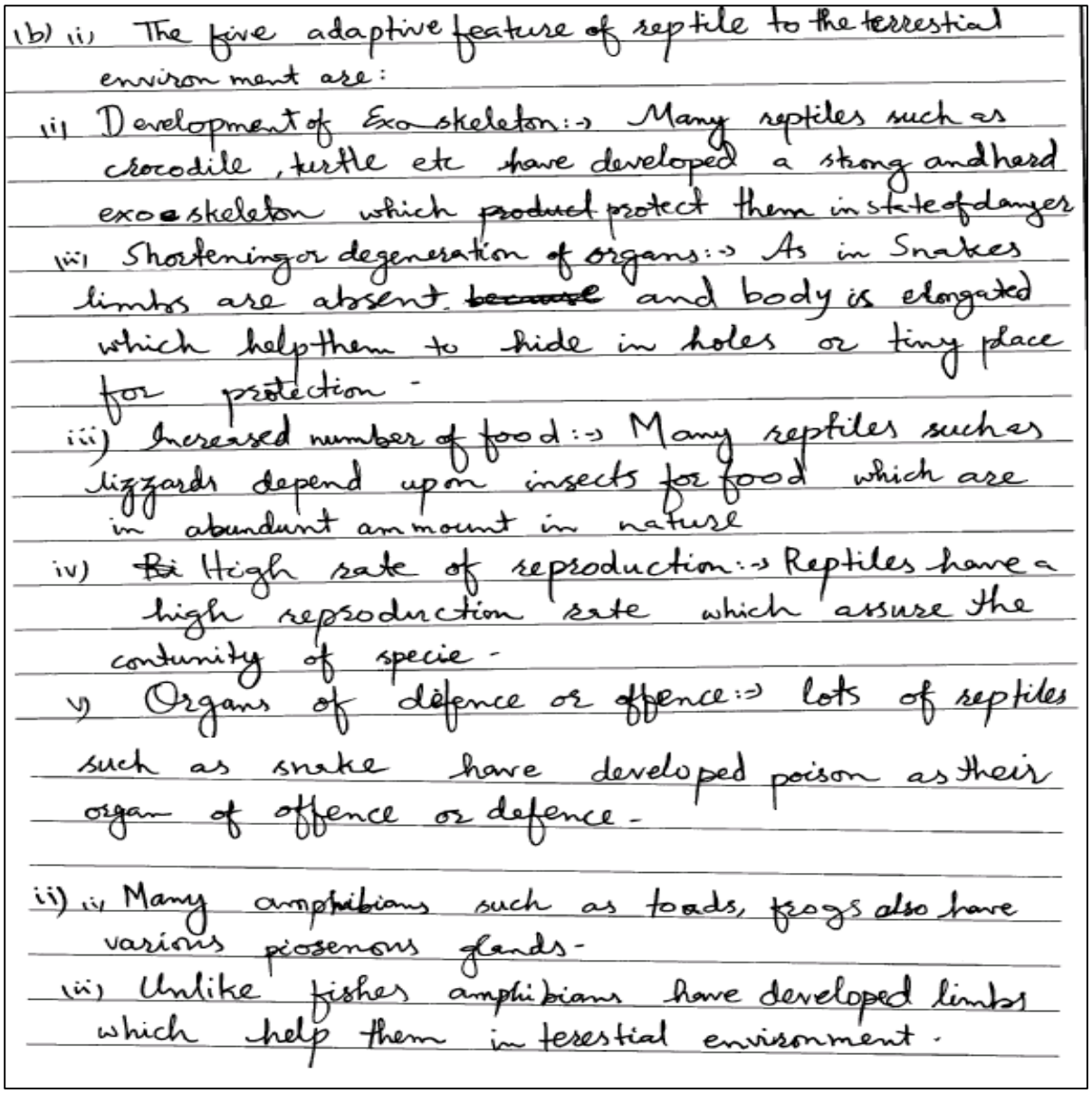
i) Reptiles adaptive features.
① Dry scaly skin which prevent them from dessication and make survival possible during harsh condition.
② No water for fertilization is required hence they can give eggs on land easily. ③ The eggs are cledoic, i-e covered with hard shell that protects from dessication or injury. ④ The embryo is covered with extra embryonic membranes "Amnion, chlorion, allantois, yolk sac" that protects the developing embryo and provide nourishment.
⑤ Pentadactylous limbs with clawed digits and scales for grip and adhesion. ⑥ Respiration is through well developed lungs. ⑦ Coloured vision to identify prey and to prevent predation.

ii) Amphibian's features:-
(a) Moist and slimy skin this can cause dryness and dessication. (b) The eggs are without shell and less amount of yolk is present, which can cause dessication of egg. (c) Require water for fertilization due to which amphibians cannot lay eggs on land they have to return to water. d) No well developed organ for respiration on land. These characters limits their changes of survival in terrestial environment.

Weaker responses provided generalised adaptive features of reptiles in the first part of the question. These responses mentioned that their circulatory/ digestive/ reproductive/ excretory and immune systems are efficient for the successful land adaptation. Furthermore, these responses identified particular examples of reptiles and described their characteristic features such as snakes reduced limb for efficient locomotion/ crocodile developed four chambered heart/ chameleon has ability to camouflage environment.

Similarly, in second part of the question, these responses provided ambiguous answers such as amphibians get unsuitable environmental conditions due to which amphibians get limited chance of survival. Moreover, these responses drafted generalised answers such as amphibians are multicellular animals/ clod blooded and heterotrophs. Few responses elaborated meaning of amphibians regardless of relating it with features which impede their chances of survival on land.

Example:



(b) i) The five adaptive feature of reptile to the terrestial environment are:

ii) Development of Exoskeleton:→ Many reptiles such as crocodile, turtle etc have developed a strong and hard exoskeleton which protect them in state of danger

iii) Shortening or degeneration of organs:→ As in Snakes limbs are absent and body is elongated which help them to hide in holes or tiny place for protection.

iii) Increased number of food:→ Many reptiles such as lizzards depend upon insects for food which are in abundant ammount in nature

iv) High rate of reproduction:→ Reptiles have a high repsoduction rate which assure the continuity of specie.

v) Organs of defence or offence:→ lots of reptiles such as snake have developed poison as their organ of offence or defence.

ii) i) Many amphibians such as toads, frogs also have various piosenous glands.

ii) Unlike fishes amphibians have developed limbs which help them in tesestial environment.

Question 11a:

The given tissues are present in the respiratory system of human beings.

- Elastic fibres
- Mucous glands
- Smooth muscles
- Ciliated epithelium

With reference to the respiratory system of human beings,

i. Name TWO organs which contain ALL of the given tissues.

ii. Describe the function of any THREE of the given tissues.

Better responses displayed sound understanding of distribution of tissues in human respiratory system by correctly identifying trachea and bronchi in the first part of the question.

In second part of the question, these responses provided relevant function of tissues. Majority of the responses written the function of mucous glands, smooth muscles and ciliated epithelium. These responses highlighted the main function tissues such as contraction and relaxation of smooth muscles alter the diameter of lumen of gas exchange organs, ciliated epithelium sweep mucous containing dust particle and mucous gland humidify the passing air. Few responses described the function of elastic fibre, as, they allow the air to be forced out of the alveoli during exhalation.

Example:

These Trachea and Bronchi of the Respiratory System contains Elastic fibres, mucous glands, Smooth muscles and Ciliated epithelium.

Mucous gland:- These are the Goblet cells which Secretes mucous in the lining of nasal cavity

Ciliated epithelium: the ciliated Squamous epithelial cells forms ciliated epithelium. It has also function in filteration of air by trapping dust particles, It also provides attachment for mucous so that it work/function properly. It is also involving in movement of mucous which is filled with dust particles towards the nose or mouth or 'pour into esophagus to release it as a waste.

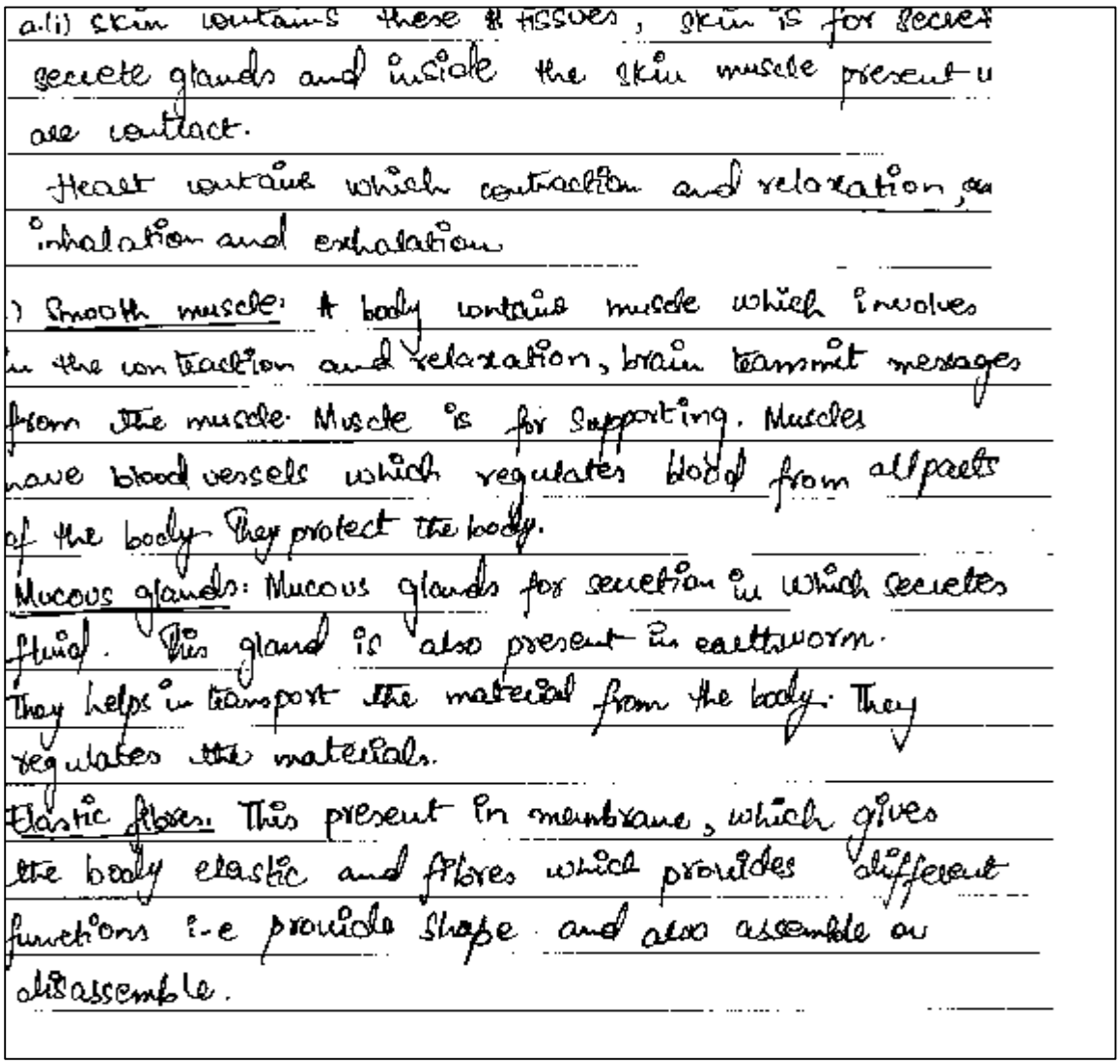
Smooth muscles:. are involved in movement of the parts of respiratory tract. They also attaches The outer layer from it to inner layer from it. Such as connective tissue /C-shape cartilage in trachea is attached with epithelium layer through muscular layer because it contain ligaments. Smooth muscles have role in expansion and contraction of lungs during inhalation and exhalation so when intercoastal muscles contracts then lungs expands so the air from environment rushes inward while when intercoastal muscles contrelan then lung contract and its surface area decrease so air rushes out.

Weaker responses exhibited candidates' poor knowledge of histology of organs involved in gaseous exchange in the first part of the question. These responses randomly named the memorised name of organs such as larynx/ pharynx/ lungs and alveoli.

In second part of the question revealed candidates carelessness in reading and understanding of the question. Majority of the candidates described the function of mucus glands and smooth muscles with reference to digestive system. Such responses described the function of mucus gland that their secretion (mucous) protects the stomach wall from damage by hydrochloric acid. Similarly, the function of smooth muscles is in movement of bolus in oesophagus/ churning of food in stomach. Moreover, function of ciliated epithelium and elastic fibres were incorrectly given. For example, ciliated epithelium trap dust particles and elastic fibres help in movement of lungs. These responses confuse with the spongy feature of lungs with elasticity.

Candidates are highly recommended to read the question completely and carefully to avoid these blunders.

Example:


a.(i) Skin contains these 4 tissues, skin is for secret secrete glands and inside the skin muscle present u are contact.

Heart contains which contraction and relaxation, as inhalation and exhalation

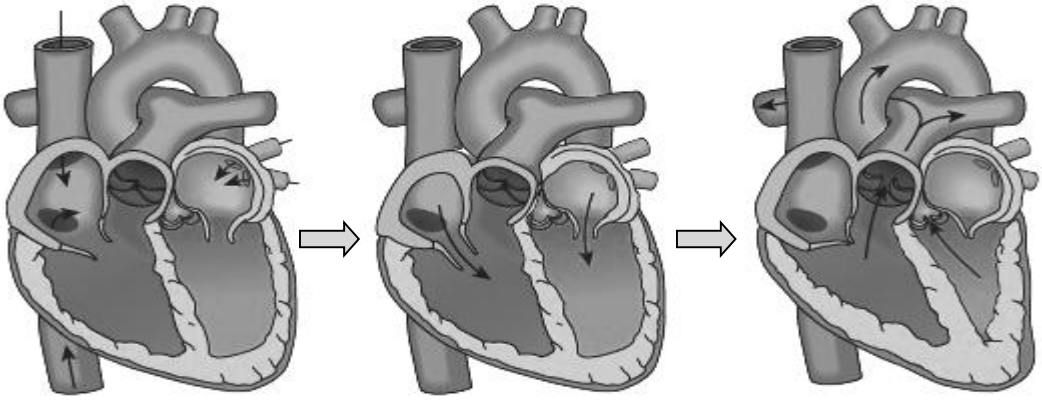
) Smooth muscle: A body contains muscle which involves in the contraction and relaxation, brain transmit messages from the muscle. Muscle is for supporting. Muscles have blood vessels which regulates blood from all parts of the body. They protect the body.

Mucous glands: Mucous glands for secretion in which secretes fluid. This gland is also present in earthworm. They helps in transport the material from the body. They regulates the materials.

Elastic fibres: This present in membrane, which gives the body elastic and fibres which provides different functions i.e provide shape and also assemble or disassemble.

Question 11b:

The given diagrams represent phases of the cardiac cycle of human beings.



Phase I **Phase II** **Phase III**

i. Describe the given phases with reference to the blood pressure changes that occur in heart chambers.

ii. Describe the regulation of cardiac cycle.

Better responses displayed wise use of diagrams in describing each phase in the first part of the question. Moreover, these responses met the demand of the question by focusing on the blood pressure change in their description. These responses related relaxed condition of atria and ventricle with lower blood pressure. Similarly, in phase II is blood pressure in the atria exceeds than ventricles as the contraction of the atrial walls forces the blood into the ventricles. In phase III, ventricles contract and increased blood within the ventricles forcing atrio-ventricular valves to close and preventing backflow of blood into atria. With the atrio-ventricular valves closed the pressure in ventricles rises.

In second part of the question, better responses clearly described the role of sino-atrial node and atrio-ventricular node in the regulation of cardiac cycle. Furthermore, these responses mentioned the role of Purkinjees fibres and bundle of hiss in their description.

Example:

b) A cardiac cycle or heartbeat is defined as the series of events which take place for the regulation and pumping of blood through the heart to other parts or organs of the body. The whole cycle consist of three phases which are described in detail on the next page :-

Phase I - The first phase is the relaxation phase of the cycle. The atrium and the ventricle of the heart are relaxed and the valves are closed. The blood moves with a high pressure in the left atrium which is brought by pulmonary vein from the lungs. Similarly the right atrium is filled with deoxygenated blood from vena cava and has pressure upto 0 mm of Hg.

Phase II - This includes the contraction of the atrium which opens the atrioventricular valves causing the blood to move with a high pressure in the ventricles. The blood from left atrium to ventricle move through bicuspid and from right atrium to ventricle through tricuspid valve. The closure of this valve produces 'lub'.

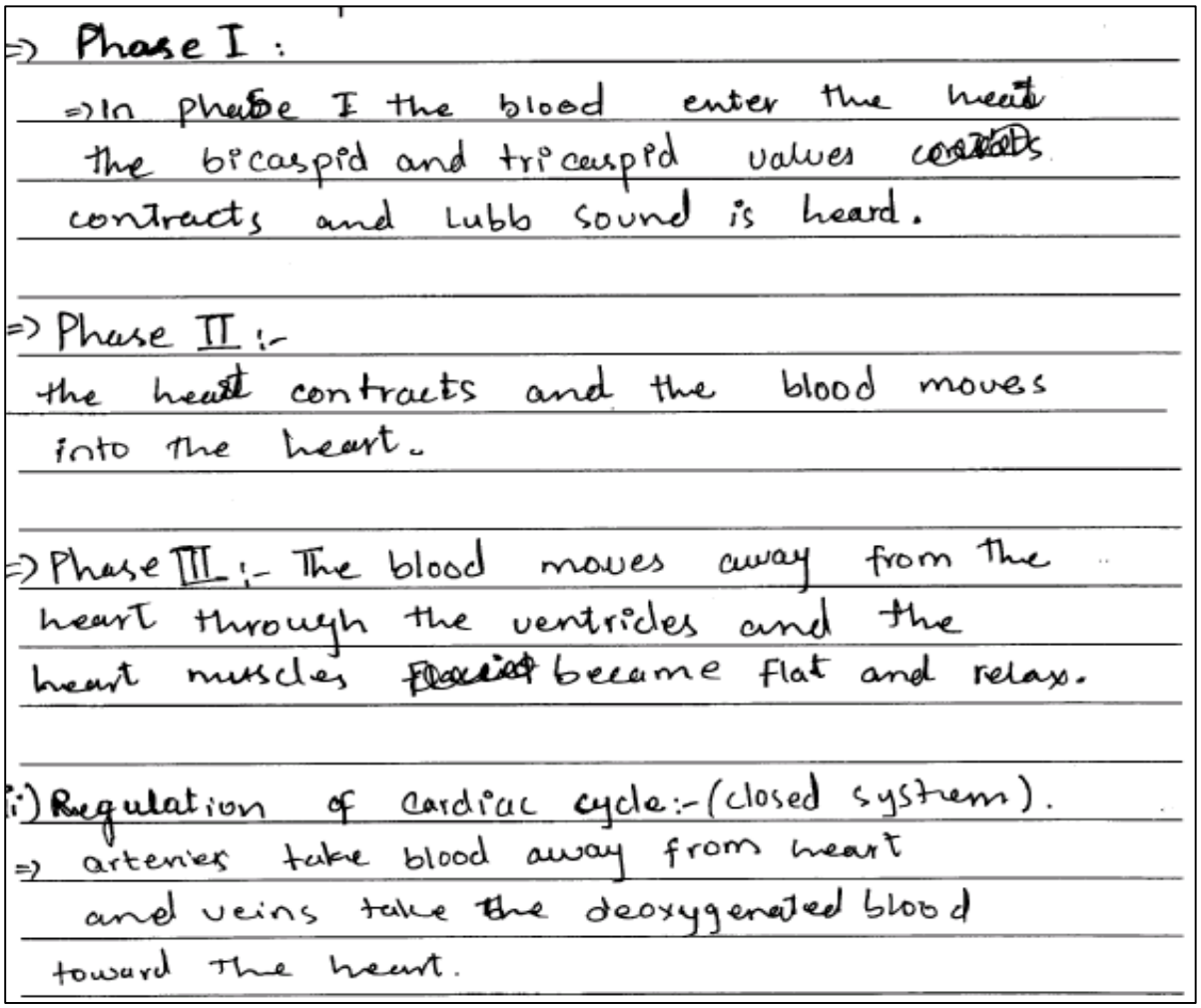
Phase III - This includes the filling and the contraction of the ventricles which opens the semi lunar valves. This pressure is comparatively lowers than atrial contraction and the blood through aortic valves move to the body and pulmonary valve to the lungs. The closure of semilunar valve produces 'dub'.

ii) The cardiac cycle is regulated with the help of electric signals originating from the sinoatrial node to atrioventricular node, AV bundles, bundles of his and purkinje fibres. The time taken for one cycle to occur is 0.8s.

Weaker responses exhibited poor understanding of diagrammatic presentation of phases of heart in first part of the question. Moreover, these responses were unable to relate the condition of heart chambers with blood pressure change. These responses drafted description of structure of heart and its function. Such as heart consist of four chambers/ right atria and left atria collect deoxygenated and oxygenated blood respectively/ right atria and right ventricle are separated by tricuspid valve/ left atria and left ventricle are separated by bicuspid valve/ pulmonary artery emerge from right ventricle and pulmonary vein emerge from left ventricle and semilunar valves are present in pulmonary trunk. Few other responses provided incomplete description by only mentioning the contraction and relaxation of heart chamber each phase without refereeing the blood pressure change. These responses were not warded full marks.

In second part of question, these responses either provided incomplete or incorrect answers. Such as Thand the responses described some steps of anaerobic respiration, electron transport chain, photosynthesis, Calvin cycle etc. Teachers are highly recommended to teach these concepts describing the mission of each process or step; this will enable students to differentiate among different concepts in bioenergetics.

Example:



=> Phase I :

=> In phase I the blood enter the heart the bicaspid and tricaspid values contracts and lubb sound is heard.

=> Phase II :-

the heart contracts and the blood moves into the heart.

=> Phase III :- The blood moves away from the heart through the ventricles and the heart muscles became flat and relax.

(i) Regulation of cardiac cycle :- (closed systems).

=> arteries take blood away from heart and veins take the deoxygenated blood toward the heart.